

Proof. **Step 1. A non-negative quadratic.** For a real parameter t define

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2. \quad (1)$$

Each summand is a square, hence $f(t) \geq 0$ for all $t \in \mathbb{R}$.

Step 2. Expanding $f(t)$. Using $(x - y)^2 = x^2 - 2xy + y^2$ we obtain

$$(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2.$$

Summing over i gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= A - 2Ct + Bt^2, \end{aligned} \quad (2)$$

where

$$A := \sum_{i=1}^n a_i^2 \geq 0, \quad B := \sum_{i=1}^n b_i^2 \geq 0, \quad C := \sum_{i=1}^n a_i b_i.$$

Step 3. Discriminant condition. A quadratic polynomial $qt^2 + pt + r$ satisfies $qt^2 + pt + r \geq 0$ for every real t iff either

* $q > 0$ and its discriminant is non-positive: $p^2 - 4qr \leq 0$; or * $q = 0$ and the resulting linear (or constant) function is non-negative for all t .

In (??) the leading coefficient is $q = B$.

- If $B > 0$, then $p = -2C$ and $r = A$. The discriminant condition yields

$$(-2C)^2 - 4BA \leq 0 \implies C^2 \leq AB,$$

i.e.

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right). \quad (3)$$

- If $B = 0$, then every $b_i = 0$. Hence $C = 0$ and (??) reads $0 \leq 0$, which is true. The same argument applies when $A = 0$.

Thus (??) holds in all cases.

Step 4. Equality case. Equality in (??) occurs precisely when the discriminant is zero (or when $B = 0$ or $A = 0$).

- *Generic case $B > 0$.* If the discriminant is zero, then $C^2 = AB$ and the quadratic can be written as

$$f(t) = B(t - \lambda)^2, \quad \lambda := \frac{C}{B}.$$

Hence

$$\sum_{i=1}^n (a_i - \lambda b_i)^2 = 0,$$

which forces each term to vanish:

$$a_i = \lambda b_i \quad \text{for all } i.$$

- *Degenerate case* $B = 0$. Then $b_i = 0$ for all i . The inequality becomes $0 \leq 0$ and equality holds for every choice of (a_i) . Since the zero vector is a scalar multiple of any vector, the condition “there exists λ with $a_i = \lambda b_i$ for all i ” is still satisfied (any λ works). An analogous argument applies when $A = 0$.

Consequently, equality holds *iff* the two vectors are linearly dependent, i.e. there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (including the degenerate case where one of the vectors is the zero vector). $\square$